J2C User Manual For I2C

Introduction

JI2C is a USB-to-I2C application for quickly reading and writing I2C device registers. Used with the JTool-I2C module, it offers advantages in interface, functionality, and performance.

Multi-device Instance Support

- Multiple devices can be used simultaneously.
- Automatic connection/disconnection upon plugging and unplugging.
- Device switching function.

Interface Design

- Flat interface, eliminating visual fatigue.
- Responsive layout, with reasonable drag/stretching.
- Large-capacity log box, smooth operation without lag or crashes.
- Automatic parameter saving.

Powerful Features

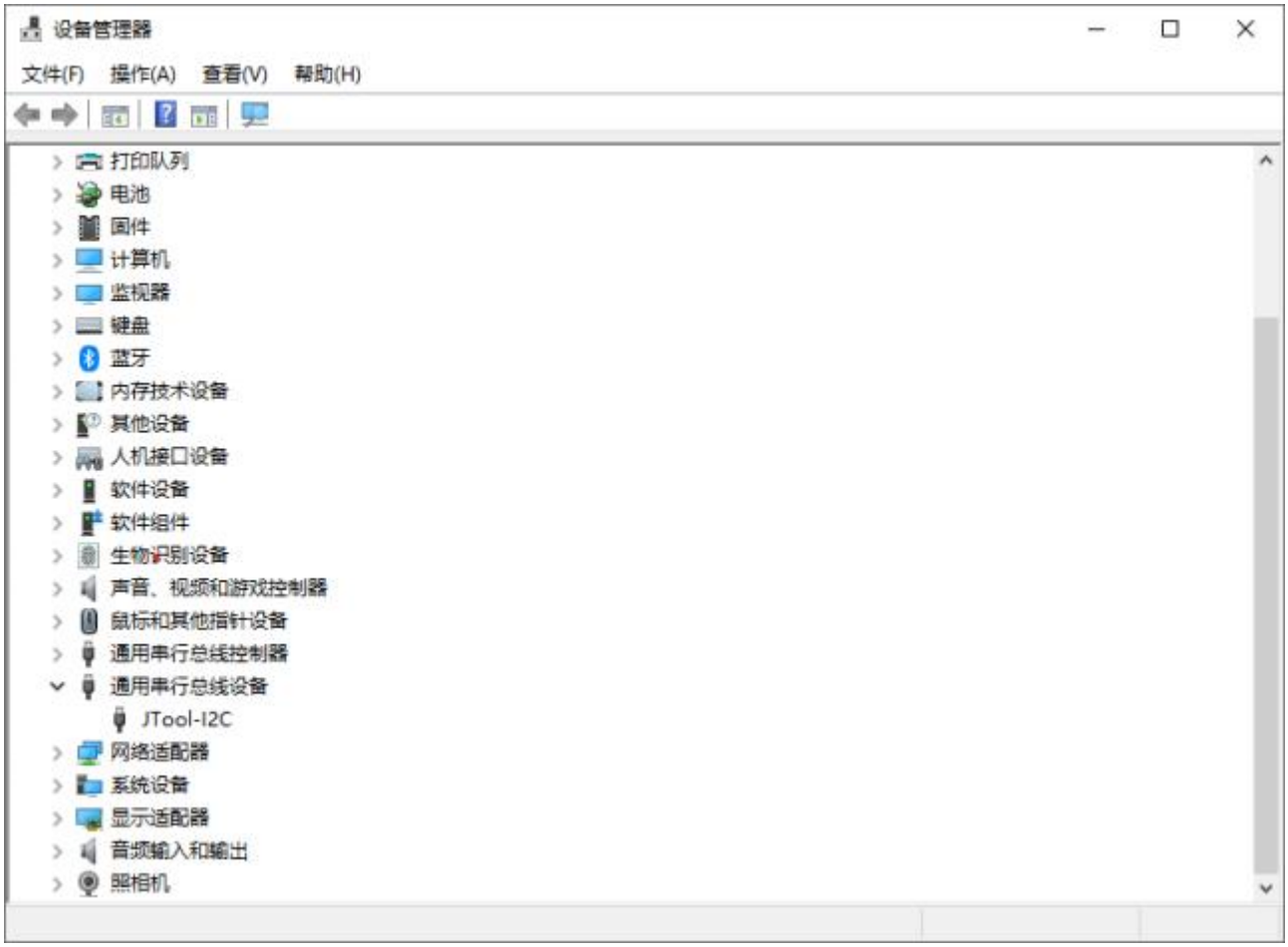
- Fast basic read/write, no need to worry about underlying protocol details such as Start, Restart, Stop, and ACK.
- Maximum single continuous read/write of 1KByte.
- Device grouping for fast switching.
- Supports interrupt-triggered and periodic-triggered operation, automatic execution.
- Supports one-click read of all data and one-click initialization.
- Supports generating real-time curves, making sensor data readily visible.
- Powerful self-developed curve control, multi-channel data display, supports arbitrary dragging, scaling, cropping, and displaying the value under mouse position.
- Supports calculation expressions, satisfying unit conversion and obtaining valid bits.
- Supports FIFO parsing.
- Supports CRC read/write verification.
- Real-time log caching and curve data caching.
- True analog slave function.

Installation

JI2C supports running on Windows 7 and later systems.

Win8 and later

No host computer installation or USB driver installation is required. Simply download the .exe file to use; the module will be automatically recognized upon insertion.



Win7

1. Requires installing .NET Framework 4.8 Go to download page
2. Manually install the JI2C WinUSB driver. Download
3. Note: The official support for Win7 systems is no longer provided. Some Win7 versions may experience incompatibility issues. **Please use a higher version of Windows.**

Instructions for Use

Inserting a Device

After inserting the JTool-I2C module, the host computer will automatically recognize and connect. When multiple devices are inserted, you can click the **ChangeDEV** menu to

switch to the desired module. In the top bar, you can set the VCC voltage, IO level (i.e., I2C bus level), and I2C speed.

The VCC voltage, IO level, and I2C speed settings will be saved internally within the module and will remain unchanged upon re-insertion.

Scan Slave Address

Click the **Scan Slave** button, and the module will check each address (0x00~0xFE) in turn to see if there is an ACK, which is used to determine which devices are on the bus. The scanned address is displayed as an 8-bit address (7-bit I2C address + 1-bit read/write bit). Use the write address, that is, the last bit is 0. (Why not use 7-bit address mode: Because 8-bit is more intuitive, most device datasheets use 8-bit, and data streams are easier to distinguish. In reality, they are all the same, just displayed differently.)

JI2C
切换设备
帮助
设备
JTool-I2C(8AC3)
VCC电压
=IO电平
IO电平
3.3V
I2C速率
1M
清空
保存文件
扫描设备
扩展
扫描到2个设备地址:
D6 FC
从机地址
00
寄存器地址类型
Reg8Bit
寄存器地址
00
读/写长度
0
读完更新
写后读取
读
写

Log Box Display Options

The six buttons on the side of the log box have the following functions:

1. Automatically clear the log box when full: Automatically clears the log box when it exceeds the display area, saving application memory during long-term operation.
2. Show read data: When selected, the read data will be displayed in the log box.
3. Show write data: When selected, the written data will be displayed in the log box.
4. Show read/write address: When selected, the read/write address will be displayed in the log box.
5. Show time: Adds a blue time marker to the log box.
6. Show line number: Displays line numbers in the log box.

The top left corner has a switch for real-time log caching to a file, suitable for saving data during long-term operation and preventing data loss due to software crashes.

Simple Read/Write

The JTool-I2C module has already encapsulated I2C Read/Write functions. We only need to input the **slave address, register address, and write data (or read length)** to read and write. In the read/write area at the bottom of the application software, set the relevant parameters, click read or write, and the log box will display the relevant results. If the "Read and Update" option is selected, the read data will overwrite the read/write box. Below is an explanation of each setting item.

JI2C
切换设备 帮助
设备
JTool-I2C(8AC3)
VCC电压
=IO电平
IO电平
3.3V
I2C速率
1M
清空
保存文件
扫描设备
扩展
23-06-10 15:08:49.226 扫描到2个设备地址.
D6 FC
23-06-10 15:08:52.422 读成功: 00 00 3F 00 00 00
23-06-10 15:08:53.880 写成功.
从机地址
D6
1
寄存器地址类型
Reg8Bit
2
寄存器地址
00
3
00 00 3F 00 00 00
4 读/写框
读/写长度
6
5
读完更新
6
写后读取
7
8
读
写

1. Slave Address: 8-bit address (with read/write bit 0). If the input read/write bit is 1, it will be automatically converted to 0 (no need to fill in the read address; if it is a read operation, the module's encapsulated function will automatically set it to 1).
2. Register Address Type: Set according to the device datasheet, referring to the length of the register address (or command).

- None: No register address required, read/write directly.
- Reg8bit: 1-byte register address or command.
- Reg16bit: 2-byte register address or command, high byte first.
- Reg24bit: 3-byte register address or command, high byte first.
- Reg32bit: 4-byte register address or command, high byte first.

3. Register Address: The register to be read or written. The length is determined by the setting in section 2. **Commands used by some devices.**
4. Read/Write Box: The read and write data boxes are one. Read data will update this text box.
5. Read/Write Length: This is linked to section 4 (read/write box). Only one of them needs to be set.
6. Update After Reading: When selected, the read data will update section 4 (read/write box); otherwise, it will not update.
7. Read After Writing: When selected, if it is a write operation, it will automatically

read again after writing to verify that the data has been correctly written to the register.

8. Read/Write Buttons: The read and write buttons are separate for greater convenience.

Grouping and Batch Reading/Write of Devices

Typically, an I2C device contains multiple registers, some used for configuration parameters and others for storing sensor data. Modifying the address and reading/writing each time would be too repetitive. Therefore, JI2C provides an extended page to summarize the information description and operation methods of each device. Added devices are automatically saved when the software exits for quick access next time. The following is an explanation of each area in the image:

1. Add/delete/select device, with a name annotation.
2. Each device has four batch operation modes, which can be switched.
3. Slave address, register address type, and interrupt type of the currently selected device. This configuration will be used for all batch lists below the selected device.
4. List display area.
5. Batch operation buttons (each of the four batch operation modes is different).
6. Operations such as adding, deleting, and moving items in the batch list.

Register Table

The purpose of the register table is to quickly view all registers inside the device and understand its current status. It has a one-click read-all function. Each register can be set to a maximum of 8 bytes, covering all types from UInt8 to Double. Descriptions can be edited for each register and viewed at any time. **The description box expands when focused and collapses when out of focus.** Additionally, entering data in the register value text box and pressing Enter allows for direct writing, facilitating quick modification of a specific register.

Initialization

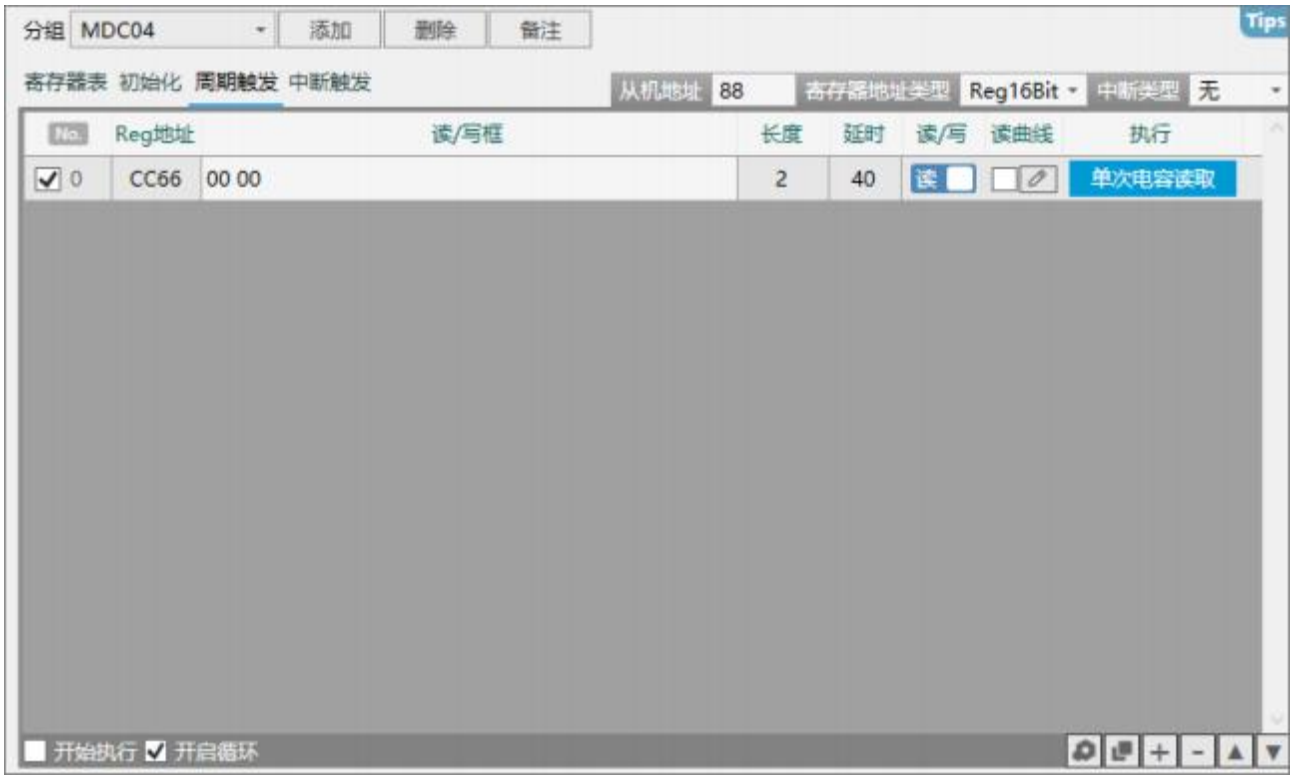
The purpose of the initialization table is to quickly configure the device to our pre-set parameters after it is plugged into the computer, enabling one-click initialization and device operation. Each initialization instruction has a separate write button; **right-clicking the button allows you to modify the notes**. Unchecking the box temporarily prevents the initialization of that instruction.

What if there are multiple different parameter configurations? There are two methods:

```
1. Use the checkboxes to select different initialization methods.
2. Add a copy of the device in the group.
```

Periodic Trigger

The periodic trigger table can execute read and write commands in the table sequentially and periodically. Each command has a configurable delay and can be configured for read/write selection. For read commands, the curve format parsing of the read data can be configured to generate real-time curves with one click.

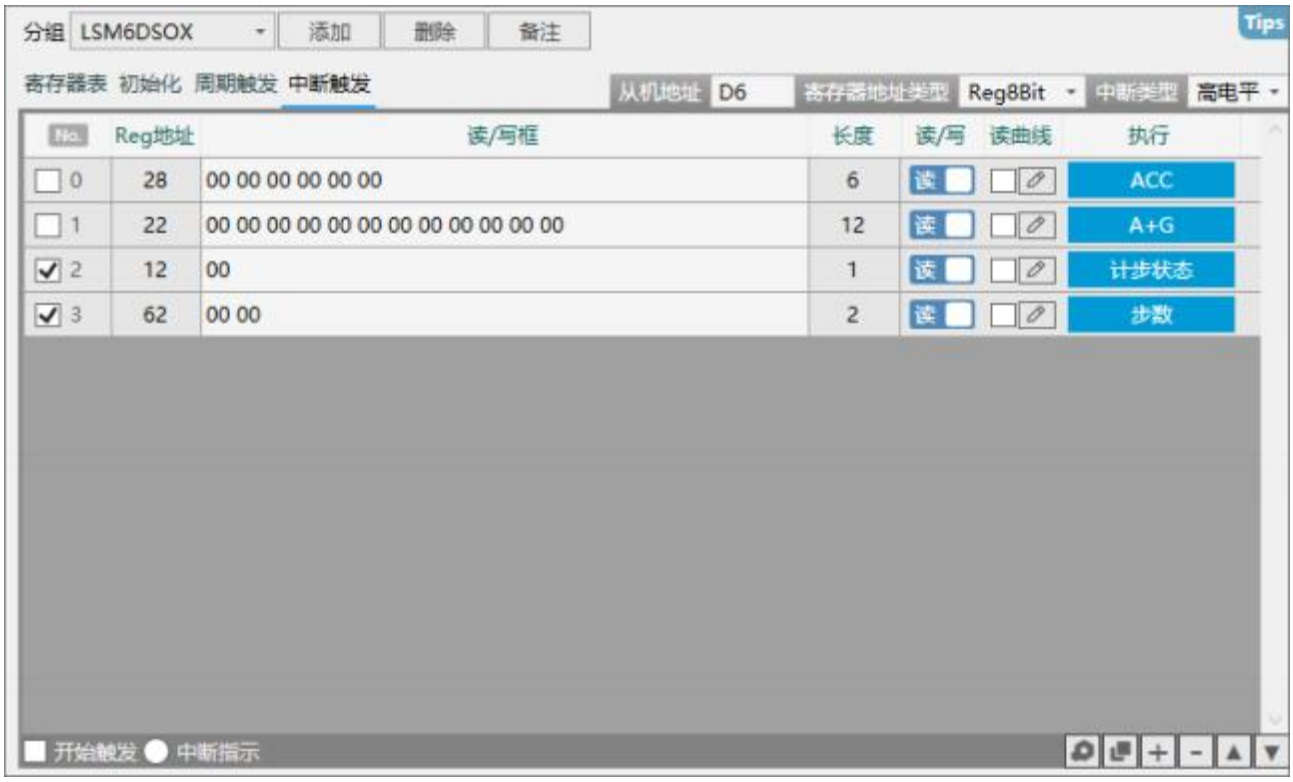


Interrupt Trigger

For most sensors, after data acquisition is complete, the host is notified to read the data via an interrupt. Therefore, the JTool-I2C module has an interrupt pin. When an interrupt is detected, it will trigger the read and write commands in the interrupt list to achieve automatic read and write. Interrupt triggering can also be configured to generate curves. Interrupt types can be configured as follows:

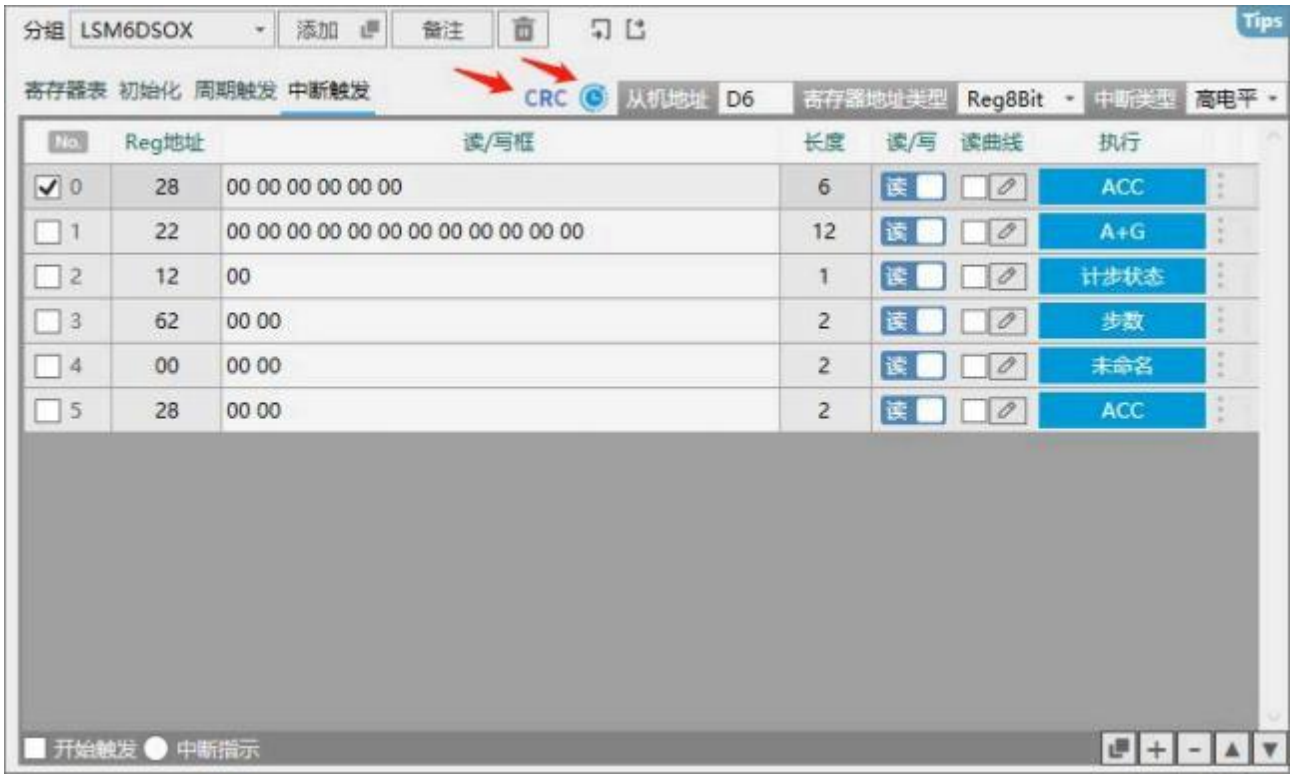
1. None
2. Rising
3. Falling
4. Edge
5. High
6. Low

Level interrupts will continuously trigger while the specified level is maintained. Currently, level interrupts trigger as frequently as once every 10ms.



CRC Check and Read Delay

CRC check and read delay options can be configured for each device group.



CRC Check

Some devices have CRC check functionality for read/write operations, and data is only accepted after successful verification during writing. Therefore, JI2C provides a byte-level CRC check option for greater convenience.

1. Enable Check
2. Select Check Range: Read/Write Content Only / All (Including Address)
3. Select Check Type, supporting the following check types:

- SUM: Accumulation
- XOR: Exclusive OR
- CRC8
- CRC8-ITU
- CRC8-ROHC
- CRC8-MAXIM
- CRC8-Custom

4. When selecting CRC8-Custom, configure the required custom parameters.
5. Verification Test: After setting the verification type, you can test whether it meets expectations.

The verification test function allows you to input a hexadecimal byte array, click "Calculate," and obtain the verification result.

After enabling verification, all read and write operations on the current device will perform verification. **Note:** The set read/write length should not include the verification byte; the software will automatically increment it by 1.

During reading, an extra byte will be read, and the verification result will be displayed in the log box.

During writing, the verification byte will be automatically calculated and added to the end of the written data.

Read Delay

Some devices, after writing the register address (or instruction) during reading, require a waiting period to allow the slave device to complete data acquisition before continuing transmission. Therefore, JI2C provides a read delay function.

1. Enable Read Delay
2. Set the Sr pre-delay time, maximum 100ms
3. Set the post-read address delay time, maximum 100ms

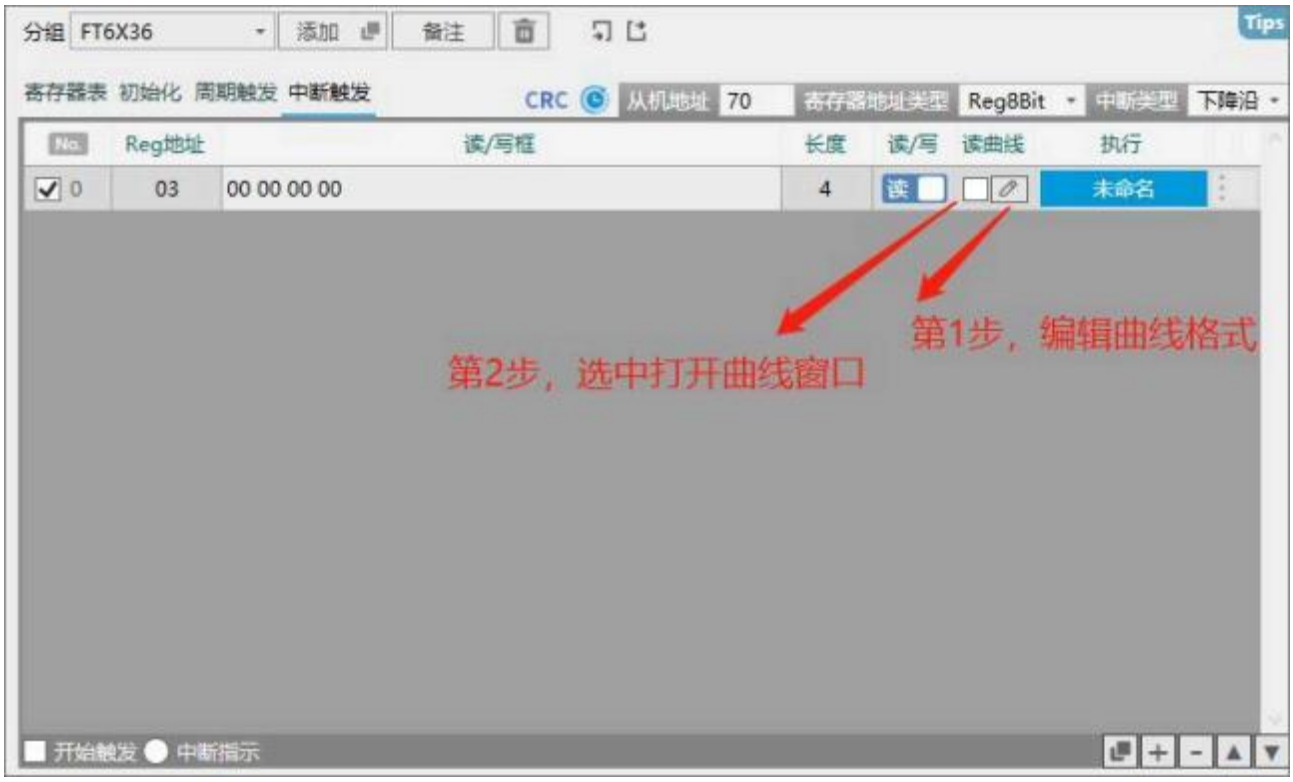
Generating Real-Time Curves

In **period-triggered** and **interrupt-triggered** modes, reading data allows viewing real-time curves.

First, the curves are based on read instructions; that is, each read instruction can generate a curve. Multiple curve windows can be opened. Selecting a curve checkbox will pop up the curve window; deselecting it will close the curve (or you can close it directly in the curve window).

Secondly, before opening the curve window, you need to edit the curve format to tell the application how to parse this data.

Editing the Curve Format


分组 FT6X36
添加
备注
Tips
寄存器表 初始化 周期触发 中断触发
CRC
从机地址 70
寄存器地址类型 Reg8Bit
中断类型 下降沿
No. Reg地址 读/写框 长度 读/写 读曲线 执行
0 03 00 00 00 00 4 读 未命名
第1步，编辑曲线格式
第2步，选中打开曲线窗口
开始触发 中断指示

曲线数据格式设定
1
通道数: 2
通道名 数据类型 高字节在前 计算公式
X坐标 UInt16 [x]&2047
Y坐标 UInt16 [x]&2047
2
3
4
5
6
使用时间作为横坐标 使能
7
FIFO/通道排列方式 按通道排列 按索引排列
排列说明
Channel 0
Channel 1
D0 D1 D2 D3 D0 D1 D2 D3
确认
取消

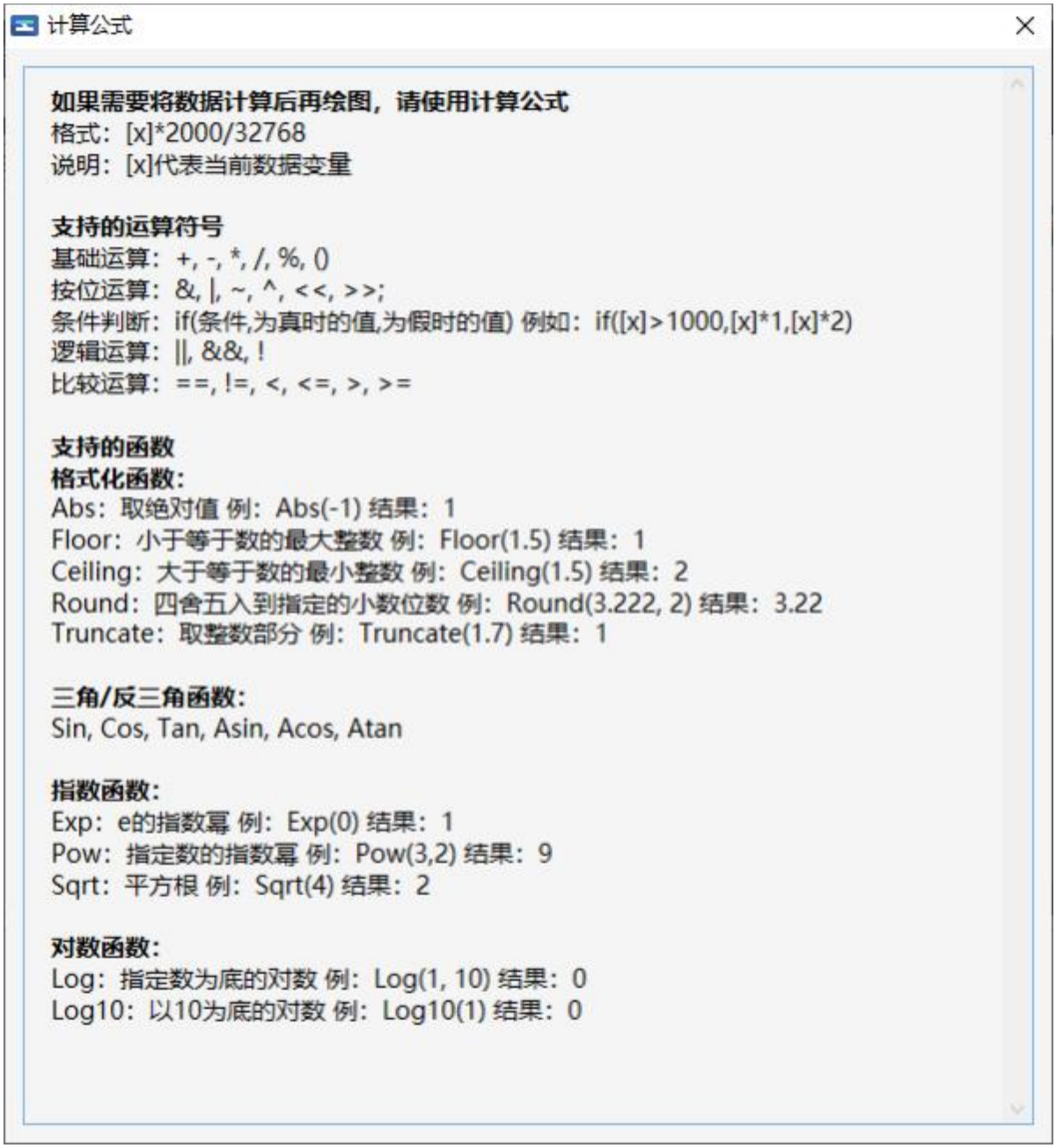
The steps for editing the curve format are as follows:

1. Enter the number of channels and press Enter, or click the Add button in the upper right corner (each read command can contain multiple bytes, and multiple channels can be read continuously at once).
2. Name the channels, which will be displayed in the curve window for easy identification.
3. Select the data for that channel Type: This option determines the number of bytes occupied by the channel and correctly parses it into the required data.
4. Select whether to use the high byte first or low byte first.
5. Select whether to use time as the x-axis index. If not, the x-axis is the data point number.
6. If you need to plot the data from this channel after calculation, please enter the expression (e.g., need to convert units, need to mask high bits, need to take a fixed number of decimal places, etc.).
7. Supports FIFO reading, i.e., each channel contains multiple data. In this case, you need to consider channel priority or index priority. **(If there is only one channel, or each channel has only one data, this option can be ignored.)**

Note: Curve generation supports multiple channels and FIFO multiple data, but the length of the calculated byte count (data type, number of channels, number of FIFOs) must match the read length of the read instruction (equal to or an integer multiple of it). Otherwise, an error will occur, and the list will be cleared. **

Calculation Expression

Click the hint button to view supported operators



Use [x] to represent the independent variable of this channel. Numbers in the expression must be represented in decimal; hexadecimal is not supported.

Tips:

1. For unit conversion, use * /
2. For offset, use + -
3. To mask high-order bits, use bitwise AND &
4. To get a fixed number of decimal places, use Round
5. Some channels are not data and are not used; you can directly multiply by 0.

Note: The calculation expression must maintain correct syntax; otherwise, an error will occur!

Curve Window

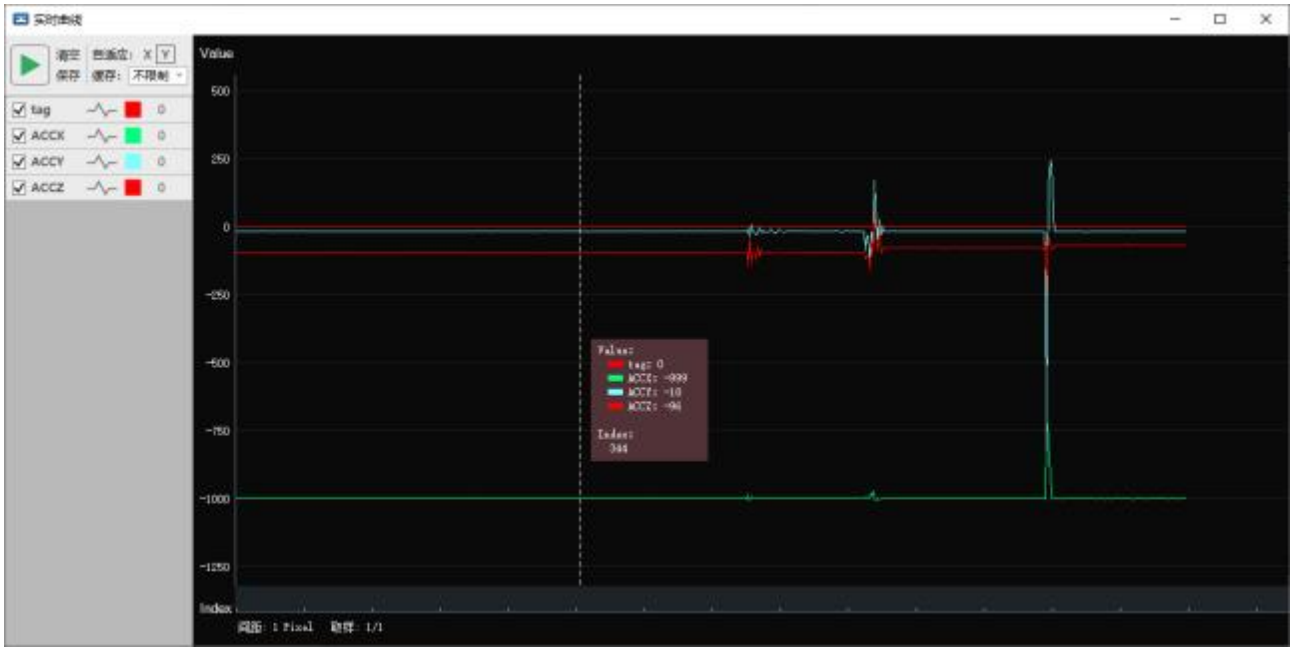
A high-performance curve control. The curve window's functions are very simple and intuitive.

Pause/Start: After pausing, data is no longer input into the curve window. At this time, you can zoom, drag, and crop.

Left-click to drag
Right-click to crop to a new window
Mouse wheel zoom vertically
Ctrl + mouse wheel zoom horizontally
Ctrl + A zoom the entire curve to fit the window size

Adaptive: Features X and Y axis adaptive options. When enabled, it will scale to the visible range on the corresponding axis.

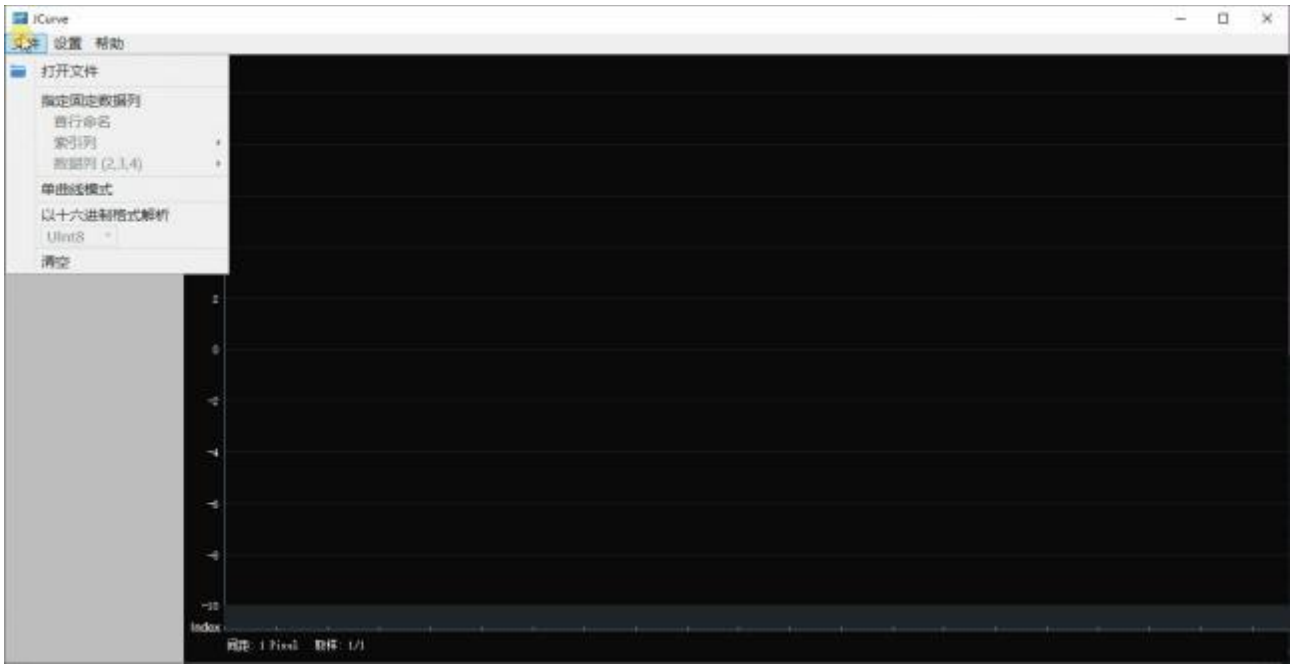
Cache: The cache here refers to the amount of data cached internally by the control. Data is cleared after reaching the set value to save memory.



The curve data is cached to a file in real time, suitable for saving data during long-term operation, avoiding software... Crash Leads to Data Loss

Static Data Curve Generation

After saving the curve data, if you want to view the curve again using this data, it is recommended to use JCurve, which supports generating static data curves by copying and pasting/opening files.



Simulating a Slave

As a slave, data needs to be pre-filled for the master to read and write at any time. The JTool-I2C module can internally simulate a slave, supporting a maximum speed of 400K,

with a maximum cache of 1KByte, and all values are 0xFF by default. Click the "Simulate Slave" button in the menu bar to enter the slave interface.

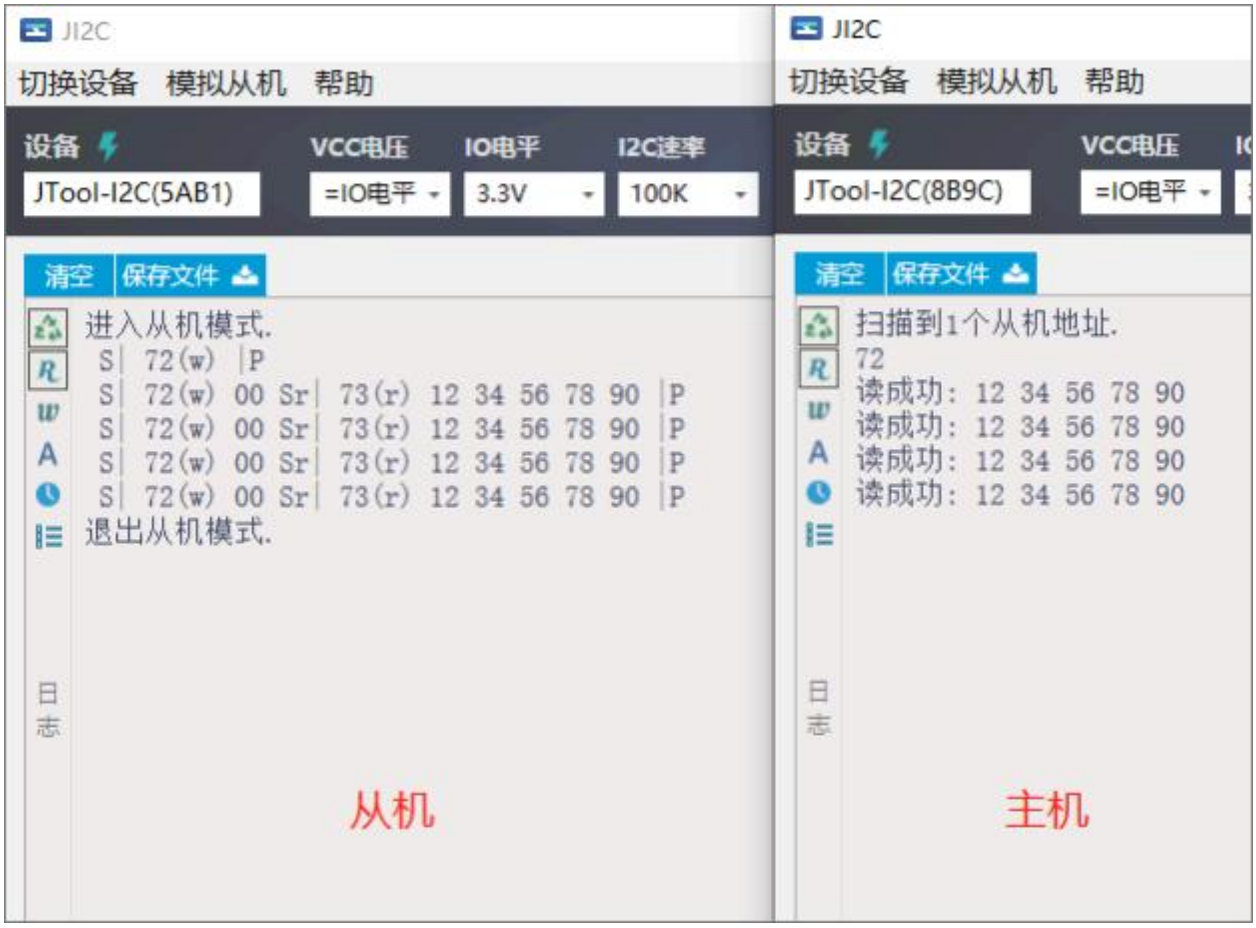
模拟从机
设置从机地址
72
响应所有地址
1
选择寄存器地址类型
Reg8Bit
2
配置内存映射
No.
Reg地址
长度
寄存器值
0
00
5
12 34 56 78 90
3
须知
4
开始模拟

1. Configure the slave address (8-bit mode), you can choose to respond to all addresses.
2. Configure the register address type, this can also be used as an instruction type.
3. Configure memory mapping, fill the specified address with the required data.

- Multiple mappings can be added.
- Addresses between mappings can be spaced out, unfilled address data is 0xFF.
- Addresses between mappings cannot overlap.
- After the mapped address exceeds 1024, it will continue to be arranged from 0.
- When the master reads, it will read the data in this list.
- When the master writes, the data in this list will be updated in real time.

4. Click "Start Simulation" to enter slave mode.
   After entering slave mode, the log box will display read and write data from the

host in real time.



1. S: Start condition
2. Sr: Repeated Start condition
3. P: Stop condition

Firmware Upgrade

The JTool-I2C module supports firmware upgrades. After the server uploads the latest firmware, it will prompt you to download and upgrade. Click Help -> About menu, download the latest firmware first, then click the Upgrade button. The module will simulate a USB drive. Drag the downloaded firmware bin file into the USB drive to complete the upgrade.

Frequently Problems

The host computer cannot run?

```
Please check if your system includes the .NET Framework 4.8.
```

Does your antivirus software or Microsoft Defender report a virus?

```
Please disable your antivirus software and Defender, or add them to the trusted
list.
```

Module not recognized?

```
1. If using a Win7 system, you need to manually install the WinUSB driver.
2. Try using a different computer.
```

Cannot find slave addresses?

```
1. If the device is already connected to another host, please disconnect it or set
 the other host's I/O port as input to avoid conflicts.
2. Check if any pins on the device require a fixed voltage level to enable the I2C
 interface.
3. Check if the configured I2C rate exceeds the device's maximum acceptable I2C
rate.
4. If using an external power supply, check if it is grounded.
```

Why does it report "NAK Received" during batch read/write?

```
Check if the slave addresses in the extended page are configured correctly.
```

Why are dozens or even hundreds of slave addresses detected?

```
Generally, scanning one address at a time is normal. Some EEPROMs, due to
partitioning requirements, have a maximum of 8 addresses.
If scanning many addresses at once, it indicates an I2C bus malfunction:
1. Incorrect wiring
2. Device not functioning correctly
3. Device's I2C interface not enabled
4. I2C speed too high
```

Is an external pull-up resistor needed?

```
No need.
```